**Problem Set for 15-May: on Kronecker Product of Matrices**

**Example** Take

- $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, - $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, - $X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}$, - $Y = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix}$. Now

1. The matrix equation $AX = Y$ coincides the linear system

$$\begin{pmatrix} a_{11} & 0 & a_{12} & 0 \\ 0 & a_{11} & 0 & a_{12} \\ a_{21} & 0 & a_{22} & 0 \\ 0 & a_{21} & 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

2. The matrix equation $XB = Y$ coincides linear system

$$\begin{pmatrix} b_{11} & b_{21} & 0 & 0 \\ b_{12} & b_{22} & 0 & 0 \\ 0 & 0 & b_{11} & b_{21} \\ 0 & 0 & b_{12} & b_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

3. The matrix equation $AXB = Y$ coincides linear system

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{21} & a_{12}b_{11} & a_{12}b_{21} \\ a_{11}b_{12} & a_{11}b_{22} & a_{12}b_{12} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{21} & a_{22}b_{11} & a_{22}b_{21} \\ a_{21}b_{12} & a_{21}b_{22} & a_{22}b_{12} & a_{22}b_{22} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}.$$

   where we write

$$\begin{pmatrix} a_{11}b_{11} & a_{11}b_{21} & a_{12}b_{11} & a_{12}b_{21} \\ a_{11}b_{12} & a_{11}b_{22} & a_{12}b_{12} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{21} & a_{22}b_{11} & a_{22}b_{21} \\ a_{21}b_{12} & a_{21}b_{22} & a_{22}b_{12} & a_{22}b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}B^T & a_{12}B^T \\ a_{21}B^T & a_{22}B^T \end{pmatrix} =: A \otimes B^T.$$

**Definition** (Kronecker product of matrices) For $P \in K^{k\times l}$, $Q \in K^{m\times n}$, define

$$P \otimes Q = \begin{pmatrix} p_{11}Q & p_{12}Q & \cdots & p_{1l}Q \\ p_{21}Q & p_{22}Q & \cdots & p_{2l}Q \\ \vdots & \vdots & \ddots & \vdots \\ p_{k1}Q & p_{k2}Q & \cdots & p_{kl}Q \end{pmatrix}.$$

Clearly, $P \otimes Q \in K^{km\times ln}$.

**Exercise 1** Verify the following statements yourself (**not a homework**):

1. $\otimes$ is an binary operation which is associative;
2. $\otimes$ is bilinear on both sides;
3. $(A \otimes B) \cdot (C \otimes D) = (A \cdot C) \otimes (B \cdot D)$.
4. $(A \otimes B)^T = A^T \otimes B^T$;
5. $\operatorname{rank}(A \otimes B) = \operatorname{rank}(A) \cdot \operatorname{rank}(B)$;
6. $\det(A \otimes B) = (\det A)^{\dim A} (\det B)^{\dim B}$.
7. $\operatorname{tr}(A \otimes B) = \operatorname{tr}(A)\operatorname{tr}(B)$.
8. Let $(\lambda, v)$ and $(\mu, u)$ be eigenpairs of $A$ and $B$, respectively. Then $A \otimes B$ has $(\lambda\mu, v \otimes u)$ as an eigenpair.

**Exercise 2** Let $J_A$ and $J_B$ denote the Jordan form of $A$ and $B$ respectively (all matrices are in $\mathbb{F}^{n \times n}$).

1. Write down the Jordan form of the linear transformation $X \mapsto AXA^T$;
2. Write down the Jordan form of the linear transformation $X \mapsto AXA$;
3. Write down the Jordan form of the linear transformation $X \mapsto AX - XA$;
4. Write down the Jordan form of the linear transformation $X \mapsto AX - XA^T$;
5. Write down the Jordan form of the linear transformation $X \mapsto AXB$.

> **Tip**
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> We assume that $A \sim A^T$ over arbitrary field, see **Exercise 7** for a proof.

**Exercise 3** Show that $\dim \ker[X \mapsto AX - XA^T] \geq n$, and explain when the equality holds.

**Exercise 4** Show that there exists $\{0, 1\}$-matrices $A$ and $B$ such that $A \otimes B$ and $B \otimes A$ are not similar.

> **Tip**
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> For square matrices $A$ and $B$, it is clear that $A \otimes B \sim B \otimes A$.

**Exercise 5 (optional)** For $A, B \in \mathbb{C}^{n \times n}$, set $A \boxplus B := A \otimes I + I \otimes B$.

1. Show that $e^{A \boxplus B} = e^A \otimes e^B$.
2. Show that $\sin(A \boxplus B) = \sin A \otimes \cos B + \cos A \otimes \sin B$.
3. Review the adjacency matrices for simple graphs (the symmetric $\{0, 1\}$-matrices labelled by edges). Explain the graph operations $A \otimes B$ and $A \boxplus B$.

**Exercise 6** Let $\varphi_i : U_i \to V_i$ $(i = 1, 2)$ be linear maps for finite dimensional vector spaces. Assume the basis of $U_i$'s and $V_i$'s are given, and so are the matrix forms of $\varphi_i$'s

1. Write down the most convenient basis of $U_1 \otimes U_2$.
2. Write down the matrix form of $\varphi_1 \otimes \varphi_2$.
3. Suppose that $U_1 = V_1$ and $U_2 = V_2$. Let $\tau : U_1 \otimes U_2 \to U_2 \otimes U_1$ be the convenient isomorphism. Write down the matrix form of $\tau \circ (\varphi_1 \otimes \varphi_2) \circ \tau$.

**Exercise 7 (optional)** Show for arbitrary field $K$, one has $A \sim A^T$. Moreover, there exists a symmetric matrix $S$ such that $S^{-1}AS = A$. Show in steps that

1. By cyclic decomposition, every matrix is similar to a block diagonal matrix wherein each block is a companion matrix of some polynomial.
2. Show that any companion matrix is similar to its transpose via a symmetric matrix (see homework for the past semesters).
3. Now conclude your proof.